Subhrajit **Prusty**

LEAD BACKEND ENGINEER

Bengaluru, India

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Summary

Backend & infrastructure engineer with 5+ years shipping and operating high-availability systems in Go and Python on AWS and Kubernetes (EKS). Build distributed services from 0 to 1 and operate them at scale: tens of millions of calls a month and ~99.5% uptime, backed by CI/CD, autoscaling, and observability that keep them reliable and cost-efficient. Deep hands-on experience with real-time voice AI, LLM-powered systems, and the GPU inference serving behind them.

Experience

Bolna

Bengaluru, India

LEAD BACKEND ENGINEER

January, 2026 - Present

- **Platform & cost:** Own the platform infrastructure behind a voice-AI product handling ~34M calls a month across ~2,400 tenants: multi-region Kubernetes (API, WebSocket, and GPU-inference clusters) scaled ~5x in six months. Own capacity planning, custom-metric autoscaling on application signals, ALB routing for WebSocket and SIP traffic, and the cloud bill, cut ~40%.
- **GPU inference:** Own the self-hosted speech-recognition platform on NVIDIA L4 GPUs on the live call path, from model-bundle lifecycle and version pinning to autoscaling on real-time stream capacity across two clouds. Built a GKE failover cluster for AWS GPU stockouts and designed a weighted per-call router across AWS, GCP, and Deepgram, the largest share of the cost cut.
- **Reliability:** Own on-call and incident management end to end: severity-classified incident reviews, runbooks linked from the alerts they serve, and command of multi-hour cross-stack incidents. Measured an MTTR/MTTD baseline that informed customer SLAs and showed detection dominated, then shipped back-tested alerts that fire earlier and page vendor outages separately.
- **Data pipelines:** Own the telemetry pipeline, shipping OpenTelemetry metrics and traces from 17+ workloads across four clusters into ClickHouse under 30-day retention, with server-span instrumentation fixed fleet-wide. Moved analytical queries off primary Postgres onto a CDC mirror and retired redundant replication; added lifecycle policies, PII-masking enforcement, and data-migration guardrails.
- **Security:** Own the security posture: an ISO-aligned, least-privilege, federated-identity model across every workload, host, CI pipeline, and developer, with per-workload IRSA, OIDC-federated CI, MFA-gated SSM-only host access, per-cluster secrets via External Secrets Operator, policy-simulation-verified IAM and RBAC, and a fleet-wide credential-rotation procedure.
- **Paved road:** Defined the new-service standard (per-service databases, IRSA, immutable arm64 images, reusable CI workflows, OpenTelemetry) and used it to take the first two internal services, an LLM evaluation harness and an agent-memory service, to production. Added Azure OpenAI as a provider and isolated tenant-level LLM failures so one customer's limits cannot degrade the shared fleet.
- **WebRTC media plane:** Took the browser-calling media plane (FreeSWITCH behind a Kamailio SIP/WSS edge) from hand-built boxes to an autoscaling fleet with golden AMIs, drain hooks, CI deploys over SSM, and alerting; validated it with a ~100-concurrent-call production load test with zero media failures.
- **Performance:** Trace infra symptoms (OOM-killed pods, stalled event loops, connection-pool exhaustion) to root causes in application code and fix both layers: cut a queue worker's memory ~95%, hardened the WebSocket fleet's health signal against event-loop stalls, and cut LLM client p99 latency ~75% under +54% traffic.
- **Post-call pipeline:** Re-architected CDR, settlement, and disposition analytics into a restart-safe, self-reconciling pipeline with atomic claim-and-checkpoint, reconciliation jobs for interrupted calls, and bounded retries, giving idempotent settlement (no double billing) and safe multi-pod scale.
- **Open source:** Built kubedump, an open-source Go CLI that snapshots live cluster state into clean, Git-trackable manifests on a daily CI schedule, the backbone for config drift tracking and audit trails across all clusters.

skit.ai

Bengaluru, India

SOFTWARE ENGINEER - PRODUCT

November 2022 - August 2025

- Engineered an LLM-powered voicebot in Go, processing millions of calls for 40+ enterprise clients; rebuilt it on the open-source Pipecat-AI framework for modularity and third-party integration.
- Designed and implemented a bidirectional audio-streaming telephony stack with gRPC for real-time interactions.
- Configured and maintained in-house + OpenSIPS infrastructure for call routing with telephony vendors like TCN and Twilio.
- Designed and implemented a flow-configuration validation service, completely eliminating config-related incidents, and optimized flow-configuration loading to cut memory usage by 80%.
- Owned the core voicebot and telephony services in production and was the de facto on-call escalation point for every voice-call incident, maintaining 99.5% uptime with Grafana, Loki, OpenTelemetry, and Prometheus.

skit.ai *Bengaluru, India*

SOFTWARE ENGINEER - SOLUTIONS *December 2020 - October 2022*

- Owned and maintained a critical internal repository, establishing code review processes and authoring contribution guidelines; significantly improving code quality and reducing technical debt.
- Developed 3rd-party API integrations, including PCI-DSS compliant Payment APIs, improving voice bot's debt collection capabilities
- Enhanced build pipelines, reducing deployment time and improving CI/CD efficiency.
- Maintained, customized and enhanced internal ML models used for real time conversational voice bots.
- Designed and implemented an automated daily reports system that simplified accessing and transforming on-prem data

racetrack.ai (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *August 2019 - March 2020*

- Using data collected by automated web-scraping, published an abstract on Personalized Hybrid-Recommender Systems using social network analysis.
- Using realtime image processing, developed a minor claims automation service for car accident processing, improving claim resolution efficiency.

ITC Infotech, Innoruption Lab (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *May 2019 - July 2019*

- Using image processing and automation, developed a metal sheet counting algorithm.
- Using facial detection and recognition, built an ID-card detail extraction system, optimizing concierge check-in process.

Education

Institute of Technical Education and Research (ITER) *Bhubaneswar, Odisha, India*

B.TECH IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY *2016 - 2020*

- CGPA : 8.81

Ispat English Medium School (IEMS) *Rourkela, Odisha, India*

INTERMEDIATE/+2 FROM ISC *2014 - 2016*

- Percentage : 84.5

Ispat English Medium School *Rourkela, Odisha, India*

MATRICULATION FROM ICSE *2014*

- Percentage : 94.17

Skills

Languages	Python, Go, Bash, SQL
Infra & Cloud	Kubernetes (EKS, GKE), Helm, Docker, External Secrets Operator, AWS (EC2/ASG, IAM/IRSA, OIDC, Secrets Manager, S3, ALB, Lambda, EFS, Budgets), GCP (GKE, Compute Engine, Filestore), NVIDIA GPU nodes, golden AMIs, GitHub Actions
ML Infrastructure	Self-hosted ASR inference (Deepgram Nova-3/Flux on L4 GPUs), GPU autoscaling, multi-cloud model serving (EKS/GKE/Modal), inference traffic routing and cost attribution, LLM provider routing (OpenAI, Azure OpenAI, Anthropic)
Observability	Datadog, Grafana, Loki, Prometheus, OpenTelemetry, Sentry, ClickHouse, ClickStack, Metabase
Backend & Messaging	FastAPI, Flask, gRPC, WebSockets, RabbitMQ, Redis, PostgreSQL, SIP/OpenSIPS, FreeSWITCH, Kamailio, Asterisk, WebRTC